

NATHAN HUBENS

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EDUCATION

Ph.D. in Computer Science

University of Mons | Institut Polytechnique Paris

Mons (BE) | Paris (FR)

Oct. 2018 - Dec. 2022

- Acceleration and compression of neural networks with parameter pruning.

M.Sc. of Engineering

University of Mons

Mons (BE)

Sep. 2016 - Jul. 2018

- Electrical engineering with multimedia and telecommunications specialization.

EXPERIENCE

FasterAI

Founder & CEO

Brussels (BE)

Jan. 2024 - Now

- Establishment of a university spin-off company specializing in neural network compression technologies.

Trusted AI Labs (TRAIL)

Research ML Scientist

Remote

Sep. 2021 - Now

- Recognized contributor in the field of Edge AI, focusing on applied research and technology transfer in resource-efficient machine learning.

ISIA Lab

Postdoctoral Researcher

Brussels (BE)

Jan. 2024 - Dec. 2024

- Applied research on neural network pruning, quantization and knowledge distillation techniques for real-world deployment.

AMD

ML Engineer

Remote

Jan. 2022 - Jul. 2022

- Model compression and speed-up with fasterai.

Creaceed

Research ML Engineer Intern

Mons (BE)

Jul. 2017 - Jun. 2018

- Image super-resolution and denoising using neural networks;
- Model compression and speed-up for mobile devices (iOS).

YEP'Tech

Executive Secretary

Mons (BE)

Sep. 2015 - Jun. 2017

- Junior Initiative providing professional experience to engineering students through projects and training.

PUBLICATIONS

FasterAI: A Lightweight Library for Neural Networks Compression

MDPI Electronics

Hubens N., Mancas M., Gosselin B., Preda M., Zaharia T.

2022

One-Cycle Pruning: Pruning ConvNets Under a Tight Training Budget

ICIP

Hubens N., Mancas M., Gosselin B., Preda M., Zaharia T.

2022

FasterAI: A Lightweight Library for Creating Sparse Neural Networks

SNN

Hubens N., Mancas M., Gosselin B., Preda M., Zaharia T.

2022

Improve Convolutional Neural Network Pruning by Maximizing Filter Variety

ICIAP

Hubens N., Mancas M., Gosselin B., Preda M., Zaharia T.

2022

Where Is My Mind (looking at)? Predicting Visual Attention from Brain Activity

Delvigne V., Tits N., La Fische L., Hubens N., Maiorca A., Wannous H., Dutoit T., Vandeborre J.-P.

MDPI Informatics

2022

Towards Lightweight Neural Animation : Exploration of Neural Network Pruning in Mixture of Experts-based Animation Models

Maiorca A., Hubens N., Laraba S., Dutoit T.

GRAPP

2022

Fake-Buster: A Lightweight Solution for Deepfake Detection

Hubens N., Mancas M., Gosselin B., Preda M., Zaharia T.

SPiE

2021

One-Cycle Pruning: Pruning ConvNets Under a Tight Training Budget

Hubens N., Mancas M., Gosselin B., Preda M., Zaharia T.

SNN

2021

An Experimental Study of the Impact of Pre-Training on the Pruning of a Convolutional Neural Network

Hubens N., Mancas M., Gosselin B., Preda M., Zaharia T.

APPIS

2020

TALKS

ICIP , One-Cycle Pruning: Pruning ConvNets Under a Tight Training Budget	Bordeaux (FR) 2022
CUTE , FasterAI: how to create small and fast neural networks	Mons (BE) 2022
SNN , FasterAI: A Lightweight Library for Creating Sparse Neural Networks	Online 2022
ICIAP , Improve Convolutional Neural Network Pruning by Maximizing Filter Variety	Lecce (IT) 2022
TRAIL Kickoff , <code>import fasterai</code> : a Library to Make Smaller and Faster Neural Networks	Mons (BE) 2022
Energy4Climate , Neural Network Compression in the time of Climate Challenge	Paris (FR) 2021
SPiE , Fake-Buster: A Lightweight Solution for Deepfake Detection	San Diego (USA) 2021
SNN , One-Cycle Pruning: Pruning ConvNets Under a Tight Training Budget	Online 2021
APPIS , An Experimental Study of the Impact of Pre-Training on the Pruning of a CNN	Las Palmas (ES) 2020

TEACHING

Faculty of Engineering , Signal Processing	Mons (BE) 2021-2023
University of Mons , Multimed'IA	Mons (BE) 2020-2022

PROJECTS

fasterai - Author

PyTorch ◊ fastai ◊ Compression

fasterai is a library for PyTorch and fastai for neural network compression (**45k+** downloads).

Kaggle - Competitor

Deep Learning ◊ Computer Vision ◊ PyTorch

Top 2% of Kaggle competitors

Medium - Technology Writer

Deep Learning ◊ Computer Vision

Writer for the *Towards Data Science* publication (**600k+** subscribers).

SKILLS & LANGUAGES

Programming: Python, Swift, C++, MATLAB;

Framework: PyTorch, fastai, TensorFlow, Keras, OpenCV, Pandas, Matplotlib, Numpy;

Language: French (native), English (level C1), Dutch (level B2), Japanese (beginner).